# FINAL REPORT - Rental Market Trends - UAE

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# EXECUTIVE SUMMARY

**Background**

The UAE's rental market is a key part of its real estate sector, with prices fluctuating based on property type, location, and market demand. This study aims to analyze rental price trends using a dataset of over 73,000 property listings, develop a predictive model for rent estimation, and identify supply-demand patterns across different cities.

**Methodology**

The dataset was analyzed using statistical techniques and machine learning models. The research involved data cleaning, correlation analysis, and predictive modeling using various algorithms.

**Key Findings**

- Location is the strongest predictor of rental price.
- Luxury areas and furnished apartments command higher rent.
- The population is skewed to limited cities hence showing uneven distribution

**Key Recommendations**

- Develop pricing strategies based on location-specific demand.
- Implement targeted marketing for underpriced areas.
- Improve rental price prediction models with additional economic indicators.

# 1. INTRODUCTION

## 1.1 BACKGROUND

Dubai's real estate market is dynamic and influenced by a number of variables, including seasonal variations, governmental regulations, and economic trends. Predictive analytics has emerged as a crucial tool for both tenants and investors looking to make well-informed decisions because real estate prices and rental rates change greatly between locations and time periods. Even while a number of research and industry reports offer valuable insights into real estate trends, there is still a lack of application of sophisticated machine learning models to accurately forecast rental prices, particularly when taking seasonal variations and transient market fluctuations into account.

(Prypco, 2024)

## 1.2 LITERATURE REVIEW

**Machine learning models**

Machine learning (ML) is a key tool in modern-day data analysis, enabling computers to learn from patterns and make predictions without explicitly being programmed. ML is usually divided into supervised learning, where models are trained on labeled data, and unsupervised learning, which identifies hidden patterns without initial labels like K-means clustering. Semi-supervised learning, which can be applied to both labeled and unlabeled data, and reinforcement learning, which is cumulative reward-based decision-making, are also significant techniques in ML. Some algorithms play a crucial role in ML applications, such as Decision Trees for classification, Naive Bayes for text classification, Support Vector Machines for complex data separation, and Principal Component Analysis for dimensionality reduction. Further, ensemble learning strategies like Bagging and Boosting provide more accurate predictions by compiling large numbers of models, while deep learning algorithms and neural networks have revolutionized AI by duplicating human processes of thinking. The increasing application of ML in applications such as recommendation systems and prediction analysis further depicts the growing role played by this technique, whose increasing significance shall further be fueled by continuing research in areas such as reinforcement learning and deep learning (Mahesh, 2018).

### The Role of Data Analytics in Dubai's Real Estate Market

The real estate sector of Dubai has taken advantage of data analytics to boost its decision-making skills. Data analytics is being used by various other stakeholders as well which include Government agencies, Property developers, and Banks. The specific data analytics that is being used include predictive modeling, sentiment analysis, and cluster analysis. (team, 2024)

Moreover, the application of data analytics in Dubai's real estate sector has led to customer service becoming better. With the knowledge of consumer behavior and preferences, companies can tailor their offerings according to market demand. This helps build a more responsive and data-driven real estate setting, further boosting market transparency and efficiency.

### Benefits of Data Analytics in Real Estate

The broader real estate industry has also benefitted from the advantages of data-driven decision-making. Data analytics provides various critical advantages that enhance operational efficiency and decision-making. Firstly, it allows accurate valuation of properties through consideration of historical sales records, location trend analysis, and macroeconomic trends. This enables both buyers and sellers to make wise decisions, reducing the likelihood of overvaluing or undervaluing properties. (Crawford, 2023)
Besides that, data analytics increases marketing efficiency through the ability to target campaigns toward specific audiences. Understanding customer behavior and preferences facilitates personalized contact, raising conversion rates. Another vital advantage is more informed decision-making, where real-time insights into data allow real estate firms to respond in a timely fashion to shifts in the market. Furthermore, the use of analytics has led to improved customer experience, as organizations can expect client needs and offer personalized solutions. Lastly, enhanced operating efficiency comes from automation and process simplification, reducing profitability and administrative complications. (Crawford, 2023)

The rental market is a crucial component of urban economy, and research on a variety of rent-determining factors indicates that location is the most important factor, with rents rising as one gets farther from business areas, transportation hubs, and amenities. zoning regulations and restrictions on the supply of homes also affect price fluctuations.

Despite these findings, it remains short of modeling rental price forecasting with real-time demand volatility and macroeconomic variables. This study aims to close this gap using machine learning algorithms and a micro-analysis of rental trends in UAE cities.

# 2. METHODOLOGY

## 2.1 RESEARCH QUESTIONS

- What are the key factors influencing rental prices in the UAE?
- How accurately can machine learning predict rental prices?
- Which locations have the highest and lowest rental demand?

## 2.2 SAMPLE

The dataset originally consisted of **73,742 property listings** across UAE cities, categorized by rent, size, furnishing, and other features.

## 2.3 DATA COLLECTION

Data was collected from Kaggle as openly available data that was in turn collected from Bayut.com.

## 2.4 DATA ANALYSIS

- Data Cleaning and Descriptive statistics to remove unnecessary rows and columns and summarize data distribution.
- Correlation analysis to identify key predictors of rent.
- Machine learning models for rent prediction.
- Visualizations to summarize rental demand and supply

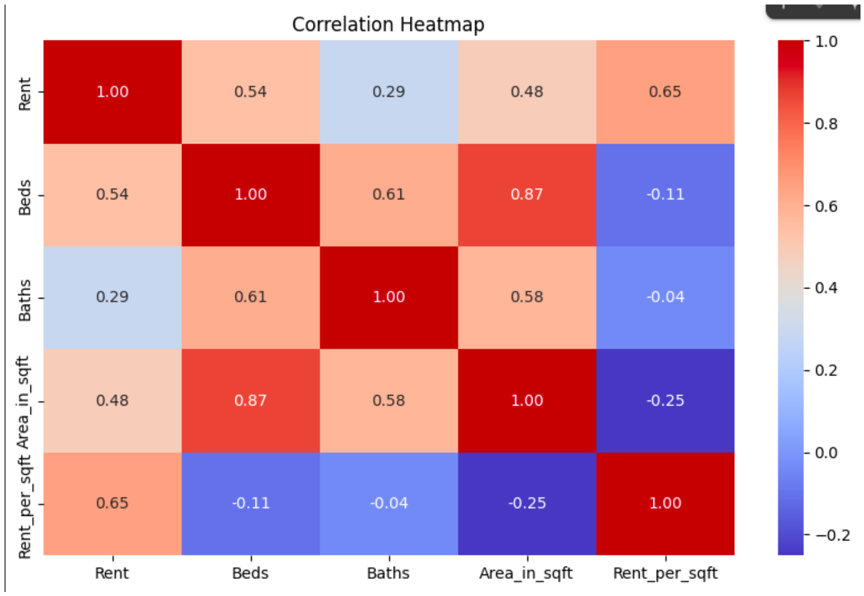
## 2.5 LIMITATIONS

- Some missing values in latitude and longitude data.
- Data is limited to only early 2024 data (majority)

# 3. RESULTS

## 3.1 Factors Influencing Rental Prices

### Correlation Heatmap




The correlation heatmap gives the following implications:

- Most of the features have positive correlation with each other
- Features like the Beds, Baths and area in sqft are negatively correlated with rent per sqft. this will be researched further in the discussion section.

## 3.2 Machine Learning Model Performance

### LINEAR MODEL

```
Accuracy:
 0.9249359964899602
Mean Squared Error of Linear Model:
 332002058.8150953
R Square of Linear Model:
 0.9249359964899602
Root Mean Squared Error of Linear Model:
 18220.92365427986
Intercept:
 175789.45420149935
Coefficients:
 [ 7.97805442e+03 -8.23223536e+03  4.83649147e+01  7.24771425e+02
  6.78503637e+03 -6.62846125e+03  4.75000020e+03  8.11156590e+02
  3.67267987e+03  1.57708082e+04 -2.06252615e+04 -8.91304808e-11
 -3.47570610e+03 -4.28086439e+03  3.37718712e+03  3.84330965e+04
 -2.58356885e+04 -1.25974080e+04 -7.58176848e+02  7.58176848e+02
 -6.32417843e+02 -2.81569024e+02  3.64621486e+02  1.46905144e+02
  2.34848537e+02  5.47730013e+01  6.22786572e+00 -3.71555482e+02
 -7.32694835e+02  5.29763460e+02  1.08610154e+03 -4.05003845e+02
  5.98343944e+03 -4.94816674e+03 -6.22732588e+03  3.55669086e+03
 -8.10681282e+03  3.01706464e+03  2.18960557e+03  4.53550493e+03]
```

**RANDOM FOREST**

```
Mean Squared Error of Random Forest Model:
 671380.1599410092
R Square of Random Forest Model:
 0.9998482043067376
Root Mean Squared Error of Random Forest Model:
 819.3779103325945
```

**GRADIENT BOOSTING**

```
Mean Squared Error of Gradient Boosting Model:
 50509823.20420365
R Square of Gradient Boosting Model:
 0.9885799818235361
Root Mean Squared Error of Gradient Boosting Model:
 7107.02632640429
```

**BAGGING**

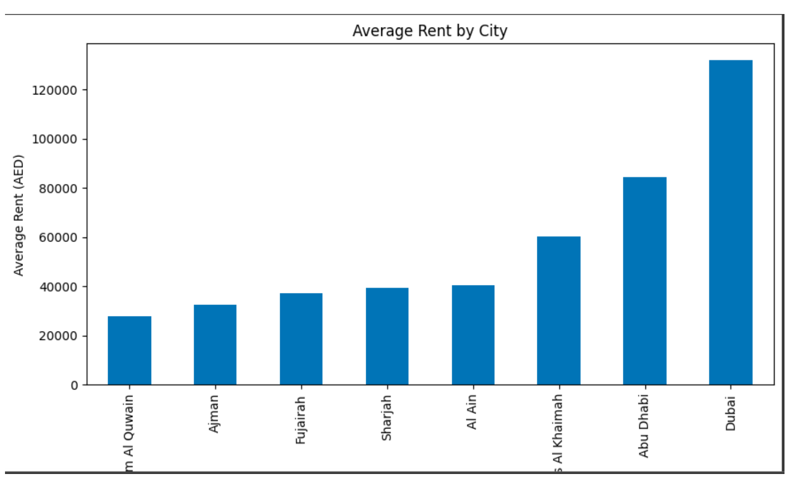
```
Mean Squared Error of Bagging Model:
 675898.1351017087
R Square of Bagging Model:
 0.9998471828151706
Root Mean Squared Error of Bagging Model:
 822.1302421768152
```

**<u>IMPLICATIONS OF THE MODELS</u>**

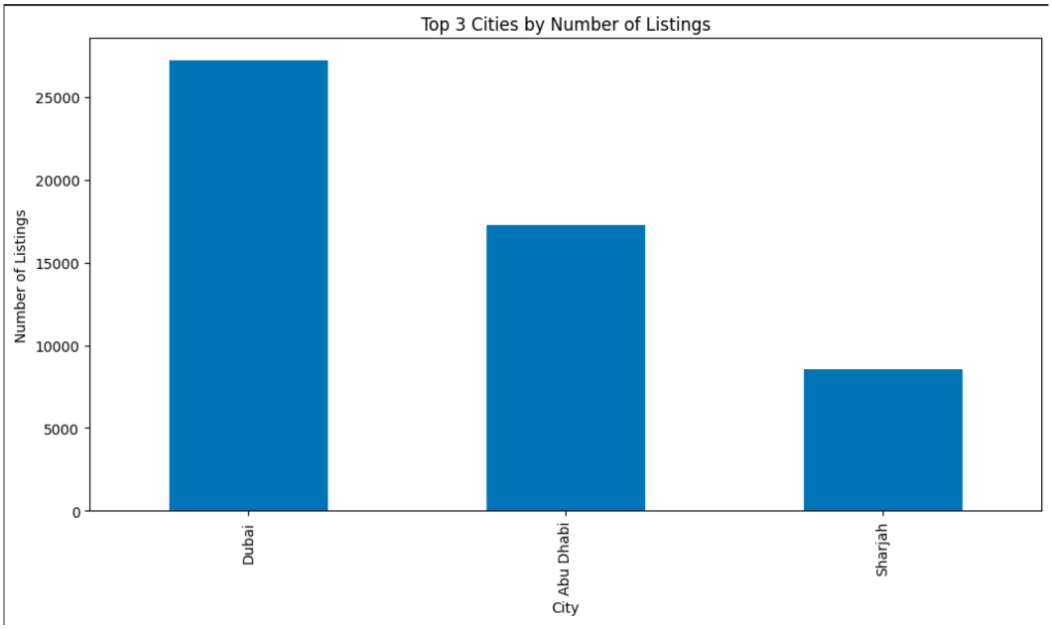
| Model | Mean Squared Error (MSE) | Root Mean Squared Error (RMSE) | $R^2$ Score or accuracy |
|---|---|---|---|
| **Linear Model** | **332002058.815 (high)** | **18220.92 (high)** | **92.5** |
| **Random Forest** | **671380.159 (very low)** | **819.317 (very low)** | **99.9848% or 0.999848** |
| **Gradient Boosting** | **50509823.204 (moderate)** | **7107.026 (moderate)** | **98.857% or 0.98857** |
| **Bagging** | **675898.135 (very low)** | **822.130 (very low)** | **99.9847% or 0.999847** |

# <u>3.3 Supply & Demand Trends</u>

**City-wise Price Trends**



**Supply vs. Demand Analysis**




Based on the graphs, it is evident that Dubai and Abu Dhabi have the highest number of property listings, as well as higher average rental prices compared to other emirates.

## GEOSPATIAL MAP ANALYSIS

Link to visualization-

file:///Users/AbdullahKhalid/Downloads/uae_rent_clusters%20(1).html

All of the high rent category(Red) areas are situated in Dubai

# 4. DISCUSSION

**CORRELATION HEATMAP**

The correlation heatmap shows that even though the majority of the features have positive correlations, features such as the number of bedrooms, bathrooms, and area in square feet are negatively correlated with rent per square foot. This can be attributed to a variety of reasons:

**1. Economies of Scale in Real Estate Pricing**

Economies of scale apply to larger properties, resulting in a reduction of cost per unit area. It simply implies the larger the property, the smaller the rent per square foot. This is the same principle as bulk purchase, where one tends to reduce the unit price by selling in bulk.
When applied to real estate, larger houses will be less expensive per square foot than smaller houses. (Lundquist, 2022)

**2. Demand Preferences for Smaller Units**

Smaller apartments, such as studio or one-bedroom apartments, are more in demand in cities. The rent per square foot rises as a result of this increasing demand. Because there may be less demand for larger units, the rent per square foot may be lower. Additionally, it has been observed that smaller apartments have considerably higher occupancy rates than their bigger counterparts, particularly in high-demand submarkets where tenants place a higher value on location than total living area. (Salmonsen, 2019)

**MACHINE LEARNING MODEL INTERPRETATIONS**

**The observations are-**

The RMSE suggests by how much of the model's rental price predictions are off by
In this case, the lowest is around 800 AED this suggests that the model is pretty close with very less error since most of the properties range from AED 50000 to 300000
The random forest has the lowest MSE (the lower the MSE the better) and hence is the best model for predicting the rental prices. However, this model might be overfitting—meaning it performs well on test data but may not generalize to new data.

Linear model has the highest MSE indicating that Linear models assume a straight-line relationship between features and rent, which may not capture complex, non-linear trends in real estate pricing.
The gradient boosting model might be a better choice for generalization than Random Forest if overfitting is a concern.
Similar to Random Forest, the Bagging model has very high accuracy with an $R^2$ of nearly 100%.
The Random Forest and Bagging models provided better predictions than Linear Regression, indicating non-linear factors in rent determination.

The overall model was limited to only the early 2024 data (reducing variability), the majority of the values being based in 2024 some being in 2023, and a small fraction of other years hence it could be a factor in the high performance level. even when testing the model, the date column was modified to only the name of the month for more efficient model performance. Hence we can suggest that only retaining the month column helps with only time-based analysis.

## BAR CHART ANALYSIS

Dubai and Abu Dhabi have the highest number of property listings, as well as higher average rental prices compared to other emirates. This trend can be attributed to several factors:

- A significant portion of the UAE's population resides in these two emirates, driving demand for housing.
- The scale of urban development and infrastructure investment is relatively higher in Dubai and Abu Dhabi, contributing to increased property values and rental rates.

The graphs show that the results confirm that **location (particularly the emirate)** is the biggest price determinant. Luxury areas maintain high prices despite supply fluctuations, while high supply in some regions reduces price growth.

## GEOSPATIAL MAP ANALYSIS

Most of the higher rents marked in red are mostly towards the new Dubai area that consist of Palm Jumeirah, Dubai Marina and Business Bay

# 5. RECOMMENDATIONS

**Apply adaptive pricing techniques**

Real estate owners and agents must adopt adaptive pricing techniques based on up-to-date supply and demand information. Rent stagnation in oversupplied areas can be mitigated by dynamically adjusting rent prices. This maintains properties competitively positioned while maximizing occupancy levels.

**Focus Investments to High-Demand Zones**

Property developers and investors should invest in areas of high demand such as Dubai and Abu Dhabi, where the highest rental prices and highest listings are recorded. This helps ensure enhanced return on investment as well as in line with directions of market demand.

**Modifying the rental price prediction models**

The data analysts and real estate specialists have to improve prediction models by incorporating macroeconomic indicators such as GDP, employment rates, and inflation rates. Current models primarily incorporate property features, but introducing wider economic indicators will increase the precision of forecasts and give stronger insights into the market.

**Develop Targeted Marketing Campaigns for Undervalued Areas**

Real estate companies' marketing teams need to target underpriced regions where rental prices are below what is expected. Personalized advertising and promotional campaigns can be used to bring in tenants to such areas to ensure rental demand is evenly distributed across various regions.

# 6. REFERENCES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# 7. APPENDICES

## APPENDIX A: DATA CLEANING, DEALING WITH OUTLIERS AND DATA SUMMARY

### DATA LOADING

```
## Load the dataset
import pandas as pd
dataset = pd.read_excel("dubai_properties 5.xlsx")
df = dataset
df.head()
#Review the dataset
df.shape
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 73742 entries, 0 to 73741
Data columns (total 17 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   Address                73742 non-null  object
 1   Rent                   73742 non-null  int64
 2   Beds                   73742 non-null  int64
 3   Baths                  73742 non-null  int64
 4   Type                   73742 non-null  object
 5   Area_in_sqft           73742 non-null  int64
 6   Rent_per_sqft          73742 non-null  float64
 7   Rent_category          73742 non-null  object
 8   Frequency              73742 non-null  object
 9   Furnishing             73742 non-null  object
 10  Purpose                73742 non-null  object
 11  Posted_date            73742 non-null  datetime64[ns]
 12  Age_of_listing_in_days 73742 non-null  int64
 13  Location               73742 non-null  object
 14  City                   73742 non-null  object
 15  Latitude               73023 non-null  float64
 16  Longitude              73023 non-null  float64
dtypes: datetime64[ns](1), float64(3), int64(5), object(8)
memory usage: 9.6+ MB
```

# DATA CLEANING

# Remove unnecesaary columns

```
df.drop('Address',axis=1,inplace=True)
df.drop('Frequency',axis=1,inplace=True)
df.drop('Purpose',axis=1,inplace=True)
df.drop('Age_of_listing_in_days',axis=1,inplace=True)
df.drop('Location',axis=1,inplace=True)
df.head()
```

| Rent | Beds | Baths | Type | Area_in_sqft | Rent_per_sqft | Rent_category | Furnishing | Posted_date | City | Latitude | Longitude |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 124000 | 3 | 4 | Apartment | 1785 | 69.467787 | Medium | Unfurnished | 2024-03-07 | Abu Dhabi | 24.493598 | 54.407841 |
| 140000 | 3 | 4 | Apartment | 1422 | 98.452883 | Medium | Unfurnished | 2024-03-08 | Abu Dhabi | 24.494022 | 54.607372 |
| 99000 | 2 | 3 | Apartment | 1314 | 75.342466 | Medium | Furnished | 2024-03-21 | Abu Dhabi | 24.485931 | 54.600939 |
| 220000 | 3 | 4 | Penthouse | 3843 | 57.246942 | High | Unfurnished | 2024-02-24 | Abu Dhabi | 24.493598 | 54.407841 |
| 350000 | 5 | 7 | Villa | 6860 | 51.020408 | High | Unfurnished | 2024-02-16 | Abu Dhabi | 24.494022 | 54.607372 |

```
df['Posted_date'] = pd.to_datetime(df['Posted_date']).dt.strftime('%B')
df.head()
```

| Posted_date |
|---|
| March |
| March |
| March |
| February |
| February |

```
df.isnull().sum()
```

| | 0 |
|---|---|
| Rent | 0 |
| Beds | 0 |
| Baths | 0 |
| Type | 0 |
| Area_in_sqft | 0 |
| Rent_per_sqft | 0 |
| Rent_category | 0 |
| Furnishing | 0 |
| Posted_date | 0 |
| City | 0 |
| Latitude | 719 |
| Longitude | 719 |

**dtype:** int64

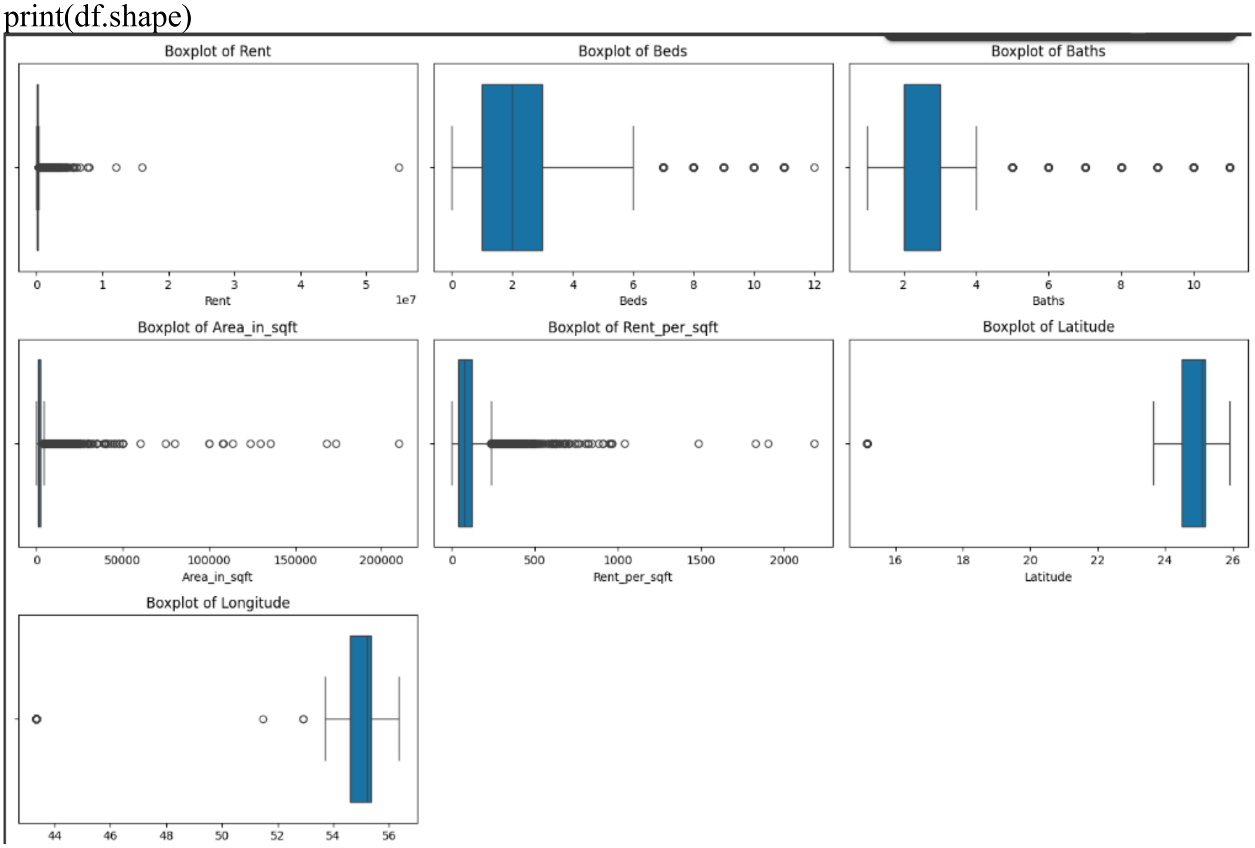
```
df["Latitude"].fillna(df["Latitude"].mean(), inplace= True)
df["Longitude"].fillna(df["Longitude"].mean(), inplace= True)
df.isnull().sum()
```

| | 0 |
|---|---|
| Rent | 0 |
| Beds | 0 |
| Baths | 0 |
| Type | 0 |
| Area_in_sqft | 0 |
| Rent_per_sqft | 0 |
| Rent_category | 0 |
| Furnishing | 0 |
| Posted_date | 0 |
| City | 0 |
| Latitude | 0 |
| Longitude | 0 |

# DEALING WITH OUTLIERS

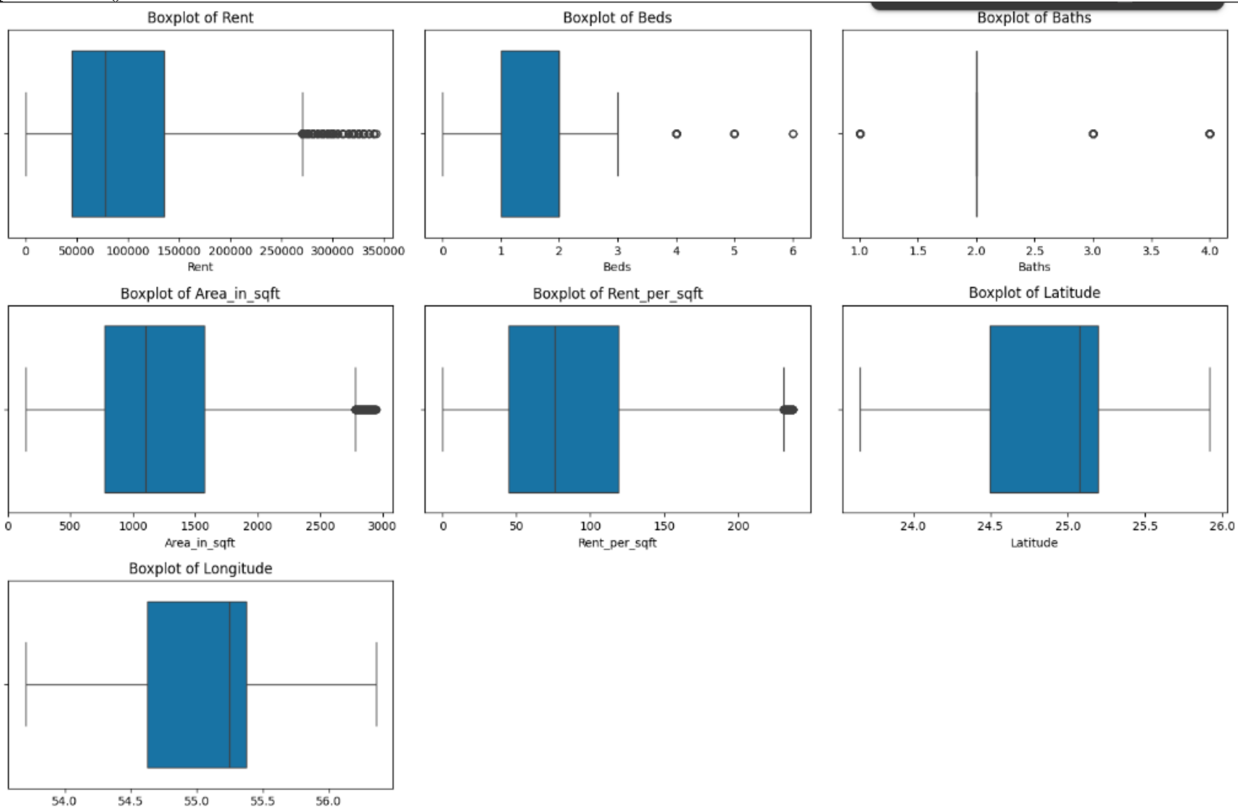
```
## Identifying Outliers for numerical columns
import seaborn as sns
import matplotlib.pyplot as plt
# Identify numerical columns
numerical_cols = df.select_dtypes(include=['int64', 'float64']).columns

# Create box plots for each numerical column
plt.figure(figsize=(15, 10))
for i, col in enumerate(numerical_cols, 1):
    plt.subplot(3, 3, i)  # Adjust subplot grid as needed
    sns.boxplot(x=df[col])
    plt.title(f'Boxplot of {col}')
plt.tight_layout()
plt.show()

## Dealing with Outliers by filtering them out
```

```
# Function to remove outliers using IQR
def remove_outliers_iqr(df, column):
    Q1 = df[column].quantile(0.25)
    Q3 = df[column].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR
    df_filtered = df[(df[column] >= lower_bound) & (df[column] <= upper_bound)]
    return df_filtered

# Remove outliers for each numerical column
for col in numerical_cols:
    df = remove_outliers_iqr(df, col)

# Display the shape of the DataFrame after removing outliers
print(df.shape)
```




```
# Create box plots for each numerical column
plt.figure(figsize=(15, 10))
for i, col in enumerate(numerical_cols, 1):
    plt.subplot(3, 3, i)  # Adjust subplot grid as needed
    sns.boxplot(x=df[col])
    plt.title(f'Boxplot of {col}')
```

```
plt.tight_layout()
plt.show()
```




# DATA SUMMARY

```
df.describe()
```

| | Rent | Beds | Baths | Area_in_sqft | Rent_per_sqft | Latitude | Longitude |
|---|---|---|---|---|---|---|---|
| **count** | 57110.000000 | 57110.000000 | 57110.000000 | 57110.000000 | 57110.000000 | 57110.000000 | 57110.000000 |
| **mean** | 97103.959534 | 1.588303 | 2.124041 | 1210.873402 | 86.191403 | 24.952229 | 55.084564 |
| **std** | 66527.145489 | 1.006573 | 0.776244 | 579.880612 | 49.946981 | 0.365442 | 0.399928 |
| **min** | 0.000000 | 0.000000 | 1.000000 | 140.000000 | 0.000000 | 23.651535 | 53.701969 |
| **25%** | 45000.000000 | 1.000000 | 2.000000 | 773.000000 | 44.736842 | 24.493598 | 54.623488 |
| **50%** | 78000.000000 | 2.000000 | 2.000000 | 1104.000000 | 75.757576 | 25.078641 | 55.245999 |
| **75%** | 134999.750000 | 2.000000 | 2.000000 | 1575.000000 | 119.133574 | 25.197978 | 55.373106 |
| **max** | 342000.000000 | 6.000000 | 4.000000 | 2946.000000 | 237.020316 | 25.920310 | 56.354219 |

# APPENDIX B: MODEL HYPERPARAMETERS

# TESTING MACHINE LEARNING MODELS

## LINEAR MODEL

```
### Modeling and Evaluation  1
###Building a linear regression model
from sklearn import linear_model
from sklearn.metrics import mean_squared_error
model1 = linear_model.LinearRegression()
model1.fit(X_train, y_train)
score1 =  model1.score(X_test,y_test)
print('Accuracy: \n', score1)
y_pred = model1.predict(X_test)
mse_lr = mean_squared_error(y_test, y_pred)
rmse_lr = np.sqrt(mse_lr)
r2_lr = r2_score(y_test, y_pred)
print('Mean Squared Error of Linear Model: \n', mse_lr)
print('R Square of Linear Model: \n', r2_lr)
print('Root Mean Squared Error of Linear Model: \n', rmse_lr)
print('Intercept: \n', model1.intercept_)
print('Coefficients: \n', model1.coef_)
```

```
Accuracy:
 0.9249359964899602
Mean Squared Error of Linear Model:
 332002058.8150953
R Square of Linear Model:
 0.9249359964899602
Root Mean Squared Error of Linear Model:
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Intercept:
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Coefficients:
 [ 7.97805442e+03 -8.23223536e+03  4.83649147e+01  7.24771425e+02
  6.78503637e+03 -6.62846125e+03  4.75000020e+03  8.11156590e+02
  3.67267987e+03  1.57708082e+04 -2.06252615e+04 -8.91304808e-11
 -3.47570610e+03 -4.28086439e+03  3.37718712e+03  3.84330965e+04
 -2.58356885e+04 -1.25974080e+04 -7.58176848e+02  7.58176848e+02
 -6.32417843e+02 -2.81569024e+02  3.64621486e+02  1.46905144e+02
  2.34848537e+02  5.47730013e+01  6.22786572e+00 -3.71555482e+02
 -7.32694835e+02  5.29763460e+02  1.08610154e+03 -4.05003845e+02
  5.98343944e+03 -4.94816674e+03 -6.22732588e+03  3.55669086e+03
 -8.10681282e+03  3.01706464e+03  2.18960557e+03  4.53550493e+03]
```

**RANDOM FOREST**

```
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error, r2_score
import numpy as np

# Initialize and train the Random Forest model
model_rf = RandomForestRegressor(n_estimators=100, random_state=42)
model_rf.fit(X_train, y_train)

# Make predictions
y_pred_rf = model_rf.predict(X_test)

# Calculate MSE, RMSE, and R²
mse_rf = mean_squared_error(y_test, y_pred_rf)
rmse_rf = np.sqrt(mse_rf)
r2_rf = r2_score(y_test, y_pred_rf)

# Print results
print('Mean Squared Error of Random Forest Model: \n', mse_rf)
print('R Square of Random Forest Model: \n', r2_rf)
```

```
print('Root Mean Squared Error of Random Forest Model: \n', rmse_rf)
```

```
Mean Squared Error of Random Forest Model:
 671380.1599410092
R Square of Random Forest Model:
 0.9998482043067376
Root Mean Squared Error of Random Forest Model:
 819.3779103325945
```

**GRADIENT BOOSTING**

```
from sklearn.ensemble import GradientBoostingRegressor
from sklearn.metrics import mean_squared_error, r2_score
import numpy as np

# Initialize and train the Gradient Boosting model
model_gb = GradientBoostingRegressor(n_estimators=100, learning_rate=0.1, max_depth=3,
random_state=42)
model_gb.fit(X_train, y_train)

# Make predictions
y_pred_gb = model_gb.predict(X_test)

# Calculate MSE, RMSE, and R²
mse_gb = mean_squared_error(y_test, y_pred_gb)
rmse_gb = np.sqrt(mse_gb)
r2_gb = r2_score(y_test, y_pred_gb)

# Print results
print('Mean Squared Error of Gradient Boosting Model: \n', mse_gb)
print('R Square of Gradient Boosting Model: \n', r2_gb)
print('Root Mean Squared Error of Gradient Boosting Model: \n', rmse_gb)
```

```
Mean Squared Error of Gradient Boosting Model:
 50509823.20420365
R Square of Gradient Boosting Model:
 0.9885799818235361
Root Mean Squared Error of Gradient Boosting Model:
 7107.02632640429
```

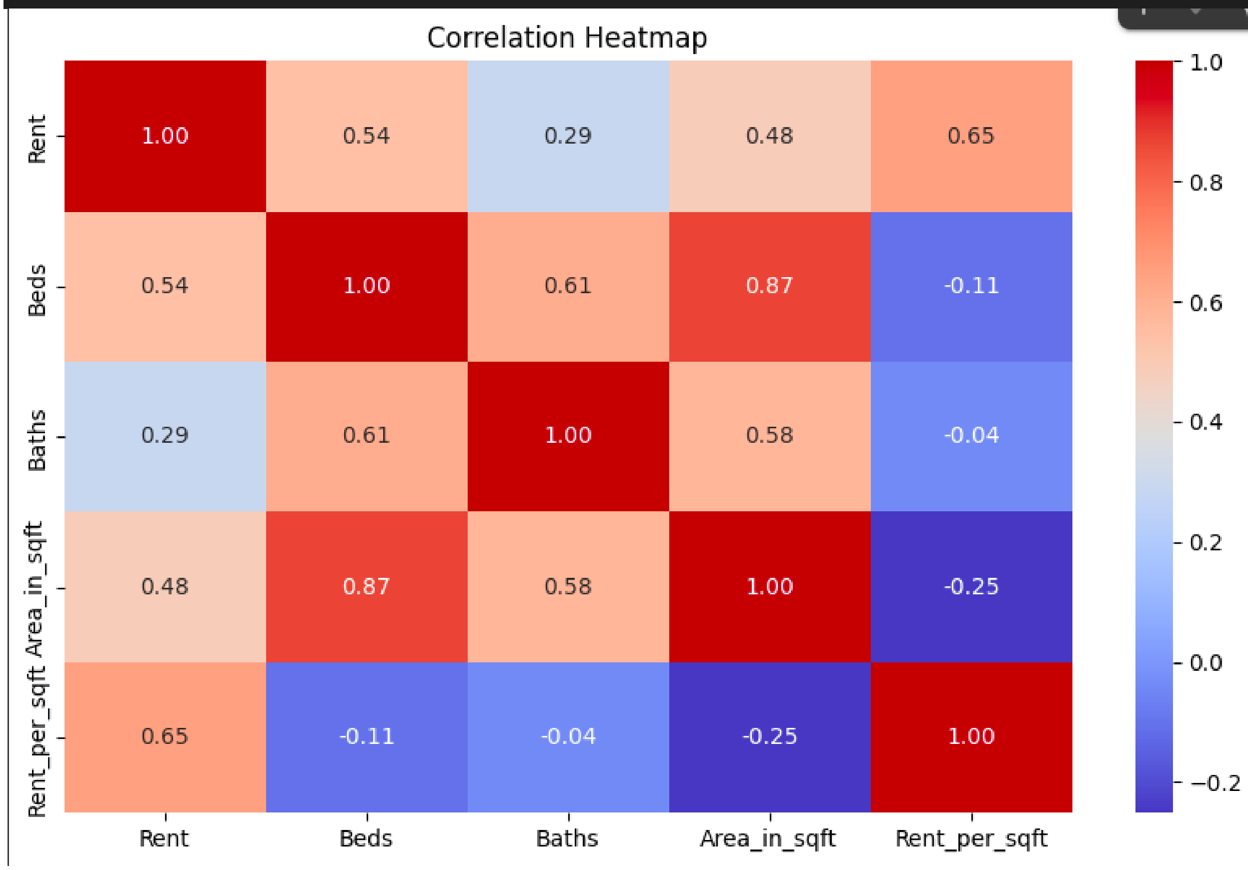
**BAGGING**

```
from sklearn.ensemble import BaggingRegressor
from sklearn.metrics import mean_squared_error, r2_score
import numpy as np

# Initialize and train the Bagging Regressor model
model_bagging = BaggingRegressor(n_estimators=100, random_state=42)
model_bagging.fit(X_train, y_train)

# Make predictions
y_pred_bagging = model_bagging.predict(X_test)

# Calculate MSE, RMSE, and R²
mse_bagging = mean_squared_error(y_test, y_pred_bagging)
rmse_bagging = np.sqrt(mse_bagging)
r2_bagging = r2_score(y_test, y_pred_bagging)

# Print results
print('Mean Squared Error of Bagging Model: \n', mse_bagging)
print('R Square of Bagging Model: \n', r2_bagging)
print('Root Mean Squared Error of Bagging Model: \n', rmse_bagging)
```

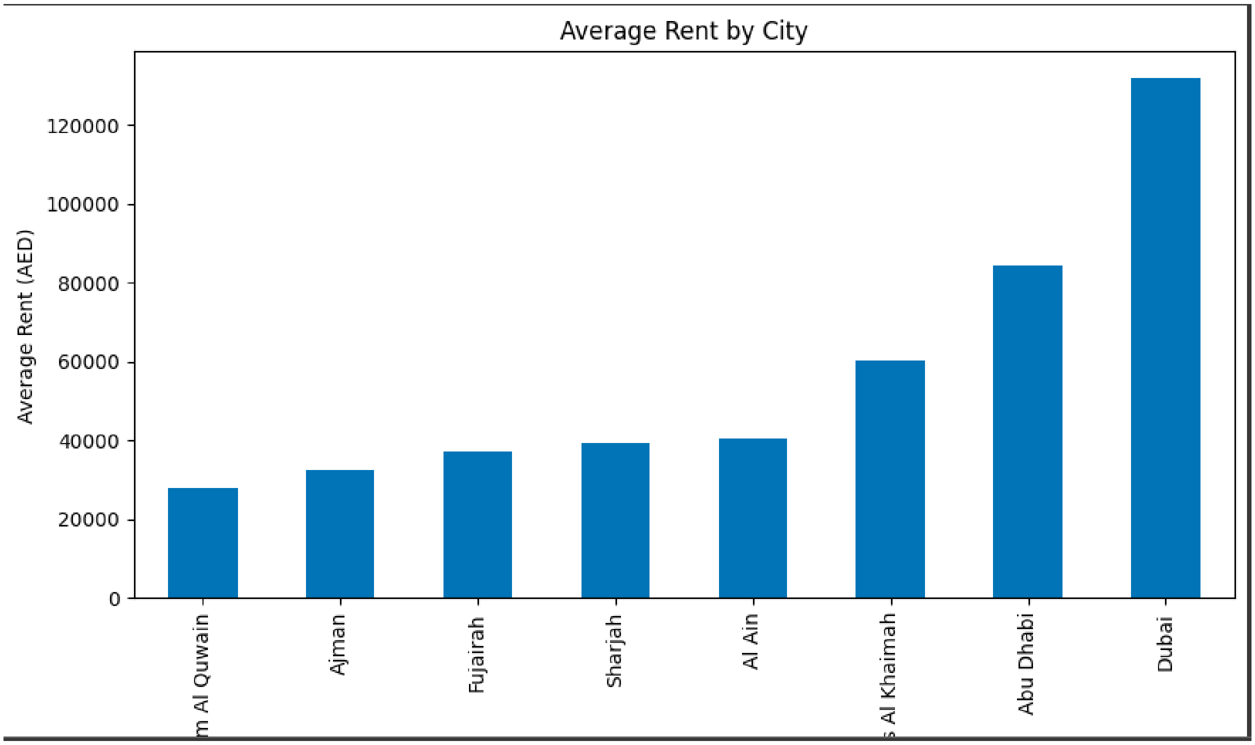
```
Mean Squared Error of Bagging Model:
 675898.1351017087
R Square of Bagging Model:
 0.9998471828151706
Root Mean Squared Error of Bagging Model:
 822.1302421768152
```

# APPENDIX C: VISUALIZATIONS

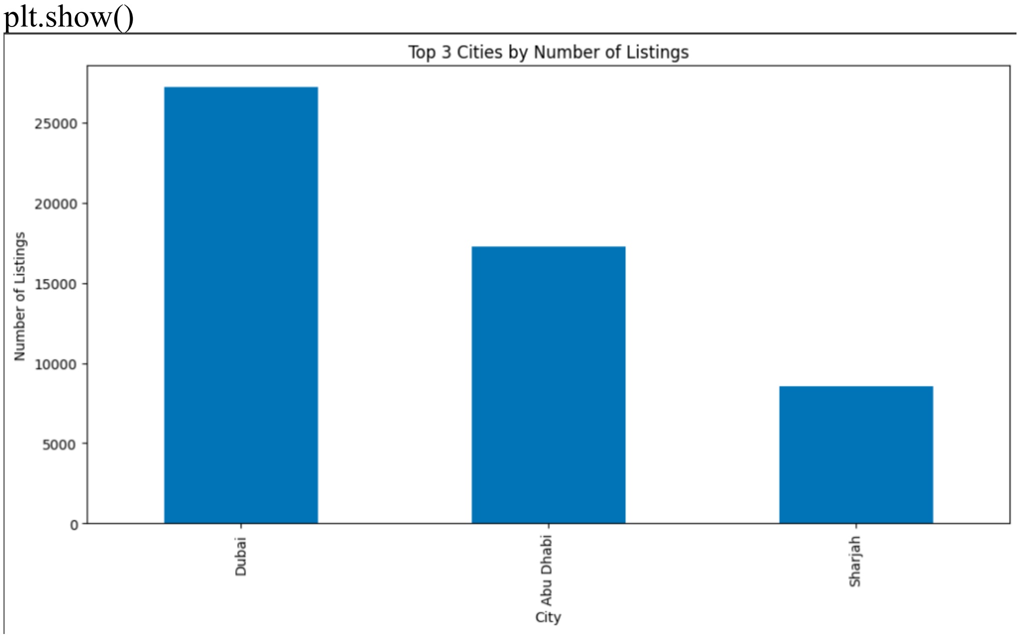
```
numerical_features = ["Rent", "Beds", "Baths", "Area_in_sqft",
"Rent_per_sqft"]
plt.figure(figsize=(10, 6))
sns.heatmap(df[numerical_features].corr(), annot=True, cmap="coolwarm",
fmt=".2f")
plt.title("Correlation Heatmap")
plt.show()
```




```
#2. City-wise Price Trends
df_city = df.groupby("City")["Rent"].mean().sort_values()
df_city.plot(kind="bar", figsize=(10,5), title="Average Rent by City")
plt.ylabel("Average Rent (AED)")
plt.show()
```




```
# 3. Supply vs. Demand Analysis
df_supply = df["City"].value_counts()
plt.figure(figsize=(12,6))
df_supply[:3].plot(kind="bar")
plt.title("Top 3 Cities by Number of Listings")
plt.ylabel("Number of Listings")
plt.show()
```



```
import folium
import pandas as pd

# Assuming 'df' is your DataFrame with Latitude, Longitude, and Rent columns
# Define bin edges for latitude and longitude
lat_bins = pd.cut(df['Latitude'], bins=10)  # Adjust the number of bins as needed
lon_bins = pd.cut(df['Longitude'], bins=10)

# Group data by latitude and longitude bins and calculate average rent and number of listings
grouped = df.groupby([lat_bins, lon_bins])['Rent'].agg(['mean', 'count'])
grouped.columns = ['Rent', 'Num_Listings']  # Rename columns
grouped = grouped.reset_index()  # Reset index to access Lat_bin and Lon_bin columns
grouped['Lat_bin'] = grouped['Latitude'].apply(lambda x: x.mid)  # Get bin center for latitude
grouped['Lon_bin'] = grouped['Longitude'].apply(lambda x: x.mid)  # Get bin center for
longitude

# Create map
uae_map = folium.Map(location=[25.276987, 55.296249], zoom_start=9)

# Plot each grouped region (latitude & longitude bins)
for index, row in grouped.iterrows():
    folium.CircleMarker(
        location=[row['Lat_bin'], row['Lon_bin']],
        radius=5,
        color='red' if row['Rent'] > 100000 else 'green',  # High vs. Low Rent
        fill=True,
        fill_color='red' if row['Rent'] > 100000 else 'green',
        fill_opacity=0.6,
        popup=f"Avg Rent: {row['Rent']:.0f} AED\nListings: {row['Num_Listings']}"
    ).add_to(uae_map)

# Save and show
uae_map.save("uae_rent_clusters.html")
```

Output

file:///Users/AbdullahKhalid/Downloads/uae_rent_clusters%20(1).html